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Efficient Glycolysis of Waste PET at Room Temperature Enabled by a Bifunctional Catalyst System and Co-solvent

Cheng Zhang, Hui He,* Fan Kang, and Hongyu Zhai

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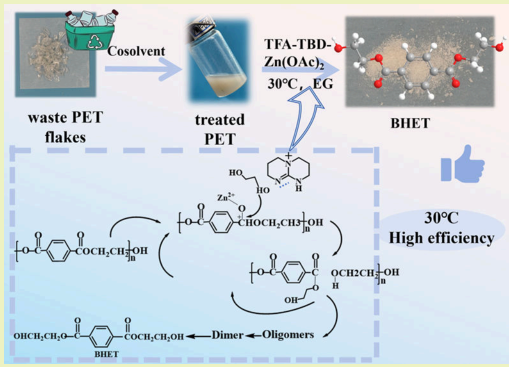
ABSTRACT: Chemical recycling represents a viable pathway for achieving closed-loop recycling of polyethylene terephthalate (PET). Although numerous chemical recycling methods for PET have been reported, depolymerization at room temperature remains rarely adopted. In this study, waste PET bottle flakes underwent glycolysis assisted by a co-solvent and a bifunctional catalyst system (TFA–TBD–$Zn(OAc)_2$), achieving quantitative PET conversion (100%) and yielding 96.1 and 90.8% bis(2-hydroxyethyl) terephthalate (BHET) under two distinct reaction regimes: 140 °C for 1.5 h and 30 °C for 3 h, respectively. Response surface methodology (RSM) identified temperature and reaction time as the most influential factors affecting glycolysis efficiency under near-ambient conditions (30−50 °C), facilitating the optimization of reaction parameters and elucidation of the reaction mechanism. This integrated approach demonstrates an effective room-temperature chemical recycling strategy for waste PET bottles, significantly enhancing the economic feasibility of industrial-scale polyester recycling.



KEYWORDS: *glycolysis, waste PET, co-solvent, bifunctional catalyst, chemical recycling*

1. INTRODUCTION

Polyethylene terephthalate (PET), one of the most widely used plastics in the polyester family, is a versatile thermoplastic polymer employed in the manufacture of bottles, textiles, packaging, and films owing to its excellent chemical, thermal, and mechanical properties, as well as its cost-effectiveness[1,2]. However, the durability of PET contributes to its environmental persistence, accounting for approximately 12% of global solid waste.[3] Although mechanical recycling currently dominates the management of PET waste, progressive degradation of polymer chains during reprocessing restricts the number of recycling cycles and compromises the quality of end-products.[4] These limitations have spurred interest in chemical recycling methods capable of regenerating materials of virgin grade. Consequently, the chemical depolymerization of post-consumer PET bottles has emerged as a promising alternative.[5] In contrast to mechanical recycling, chemical depolymerization enables the controlled breakdown of macromolecular chains of PET into fundamental monomeric units, specifically yielding either terephthalic acid (TPA) through hydrolytic cleavage[6] or bis(2-hydroxyethyl) terephthalate (BHET) via glycolysis.[7] Glycolysis has emerged as a particularly promising approach for chemical recycling due to its (i) straightforward operational requirements, (ii) mild reaction conditions, and (iii) seamless integration potential with conventional facilities for polyester manufacturing.[8,9] Significant advancements have been achieved in catalytic systems for the glycolysis of polyesters in recent years, ranging from conventional homogeneous catalysts (e.g., zinc acetate and ionic liquids)[10,11] to sophisticated heterogeneous systems, particularly metal–organic frameworks (MOFs)[12,13] and tailored mesoporous oxides.[14,15] Catalysts for the glycolysis of PET are primarily classified into homogeneous and heterogeneous systems. While heterogeneous catalysts facilitate easier separation and recyclability, they often encounter resistance to mass transfer. Conversely, homogeneous catalysts, such as metal acetates, offer superior performance due to their dispersion at the molecular level. Their primary advantages include enhanced kinetic efficiency, milder reaction conditions, and the elimination of interfacial barriers, ensuring rapid and complete depolymerization.[16−20] Mechanistically, these catalytic systems can be classified into three distinct categories: (1) Lewis acid catalysts,[16,17] (2) Lewis base catalysts,[18] and (3) hydrogen-bond donor catalysts.[19,20] The catalytic mechanism of Lewis acid catalysts involves coordination between Lewis acidic sites and carbonyl oxygen atoms, inducing a partial positive charge on the carbonyl carbon. This electronic activation significantly enhances the susceptibility of the ester bond to nucleophilic attack by ethylene glycol (EG). In contrast, Lewis base and

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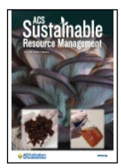

hydrogen-bond donor catalysts primarily modulate the electronic properties of EG through either proton abstraction or hydrogen-bond formation. These interactions not only increase the electron density on the oxygen atoms of EG but also markedly enhance their nucleophilicity toward the cleavage of ester bonds.[21,22] Among nitrogen-based organic catalysts reported in the literature, 1,5,7-triazabicyclo[4.4.0]dec-5-ene (TBD), 1,8-diazabicyclo[5.4.0]undec-7-ene (DBU), and 4-dimethylaminopyridine (DMAP) have demonstrated exceptional catalytic performance.[8,20,23,24] Notably, complete depolymerization of PET can be achieved within approximately 60 minutes using TBD as a catalyst under a N_2 atmosphere at 190 °C with excess EG.[23] However, harsh reaction conditions (>180 °C) are typically required to achieve satisfactory rates of depolymerization in these conventional catalytic systems.[21,22]

To address these limitations, recent research has focused on developing bifunctional catalysts for the synergistic catalysis of the glycolysis reaction of PET.[16,18] These advanced catalytic systems operate via concerted hydrogen-bonding interactions between the molecular orbitals of the catalyst and the electron clouds of the substrate. Bifunctional catalysts, typically engineered through either composite active sites or architectures of porous support, integrate complementary Lewis basic sites (e.g., hydroxyl or amine groups) and Lewis acidic centers (e.g., metal oxides or MOF nodes).[25,26] For instance, pronounced catalytic synergy is achieved in the ZnO/nanosilica composite system, where Zn^{2+} centers act as Lewis acids to polarize ester groups, while surface oxygen vacancies function as Lewis bases to facilitate nucleophilic attack.[27] Carboxylate ligands have been demonstrated to play a pivotal role in the catalytic depolymerization of waste PET. Accordingly, we selected the representative trifluoroacetate anion and combined it with Zn^{2+} and either TBD or DBU to construct bifunctional catalysts that simultaneously exhibit Lewis acid and Lewis base characteristics. These catalysts were then employed in the glycolysis of waste PET using ethylene glycol.[28]

Despite significant progress, the persistent reliance on elevated reaction temperatures remains a fundamental limitation in the glycolysis of PET.[29] Current strategies for temperature reduction often involve conditions of high-energy input to achieve multiscale optimization. For example, microwave-assisted reaction intensification has been shown to reduce energy consumption by 40% while maintaining high rates of depolymerization.[8,30] Alternatively, the introduction of co-solvents has proven effective in accelerating reaction kinetics while suppressing undesirable side reactions.[31] Under homogeneous conditions, complete depolymerization of PET can be achieved within approximately 3 hours at 180 °C.[32] However, energy-intensive methods such as microwave irradiation or supercritical fluid technology[33,34] are required when conducting glycolysis reactions below 100 °C, often accompanied by longer reaction times. While various intensification techniques have been implemented to improve glycolysis efficiency, achieving effective depolymerization below 100 °C remains a substantial challenge.[35,36] Notable breakthroughs in low-temperature glycolysis include the work by Lu et al.,[29] who achieved a 90% yield of BHET at 90 °C using acetonitrile as a co-solvent and $ChCl/Zn(OAc)_2$ as a catalyst during 12-hour reactions. The swelling effect of acetonitrile induced surface cracks and defects in PET, accelerating the depolymerization reaction. Furthermore, acetonitrile was found to significantly promote the degradation of oligomers into BHET.[29] More recently, innovative solvent–catalyst systems employing 1,4-dioxane or tetrahydrofuran as co-solvents have enabled the glycolysis of PET at temperatures as low as 65 °C. These developments suggest that the rational design of novel catalytic systems incorporating co-solvent-assisted mechanisms holds great promise for the efficient chemical recycling of waste PET bottles under mild or even ambient temperature conditions.

In this study, an appropriate co-solvent was selected to assist in the dissolution of waste PET bottle flakes. Subsequently, a multiple hydrogen-bond catalyst, TFA–TBD–$Zn(OAc)_2$, was prepared and employed for the glycolysis of waste PET based on the assistance of the co-solvent. Response surface methodology was used to explore the factors affecting the glycolysis efficiency of waste PET, and kinetic studies were conducted. The activation energy of the glycolysis reaction catalyzed by TFA–TBD–$Zn(OAc)_2$ was determined, and the reaction mechanism was elucidated. This study achieves efficient chemical recycling of waste PET bottle flakes at ambient temperature, significantly improving monomer yields while enhancing the economic viability of large-scale industrial polyester recycling.

2. EXPERIMENTAL SECTION

2.1. Materials

Waste PET flakes (3 mm × 3 mm) were supplied by an industrial manufacturer. Analytical grade solvents, including dimethyl sulfoxide (DMSO), dichloromethane (DCM), *N*,*N*-dimethylformamide (DMF), and *N*-methyl-2-pyrrolidone (NMP), 1,3-dimethyl-2-imidazolidinone (DMI), and 1,1,2-trichloroethane (TCE), were purchased from FuChen Chemical Reagent Co., Ltd. The catalysts 1,5,7-triazabicyclo[4.4.0]dec-5-ene (TBD), 1,8-diazabicyclo[5.4.0]undec-7-ene (DBU), and zinc acetate ($Zn(OAc)_2$) were obtained from Macklin Biochemical Co., Ltd. (Shanghai, China). All chemicals were used as received without further purification.

2.2. Treatment of PET Waste Flakes

Waste PET bottle flakes were crushed three times in a crusher for 3 minutes. After crushing, the size of the waste PET bottle flakes was significantly reduced (approximately 1 mm × 1 mm), which is conducive to increasing the reaction surface area with the co-solvent.

2.3. One-Step Preparation of Catalysts

The bifunctional catalysts were synthesized by a stoichiometric combination of trifluoroacetic acid (TFA), TBD, and $Zn(OAc)_2$ with a molar ratio of 3:1:1 in a round-bottom flask (seen as Figure 1a). The mixture was allowed to react under continuous stirring at 120°C for 3 h until formation of a homogeneous transparent solution. Then, the resulting products (TFA–TBD–$Zn(OAc)_2$) were dried in a forced-air oven at 70°C, followed by final dehydration under vacuum (−0.1 MPa) to remove trace moisture. TFA–$Zn(OAc)_2$–EG and TFA–TBD were prepared by the same method as described above.

2.4. Depolymerization of Waste PET Flakes Assisted with Co-solvents

$m_{0,PET}$ of crushed waste PET flakes, a certain mass of co-solvents (such as DMF, NMP, DMSO, etc.), and EG were added into a round-bottom flask. Waste PET bottle flakes were dissolved at a predetermined temperature for a set duration until a milky solution was obtained (referred as treated PET). Next, the catalysts were added into the reaction system, and the glycolysis reaction was carried out at a certain temperature (seen in Figure 1b). After the reaction was completed, the reaction mixture was filtered to remove the unreacted PET fragments (mass recorded as $m_{1,PET}$) and then an excess amount of 80 °C distilled water was introduced into the mixture. The second filtration was carried out. After that, the filtrate was placed in the refrigerator at 0 °C for 12 hours. Finally, the white crystals (BHET)

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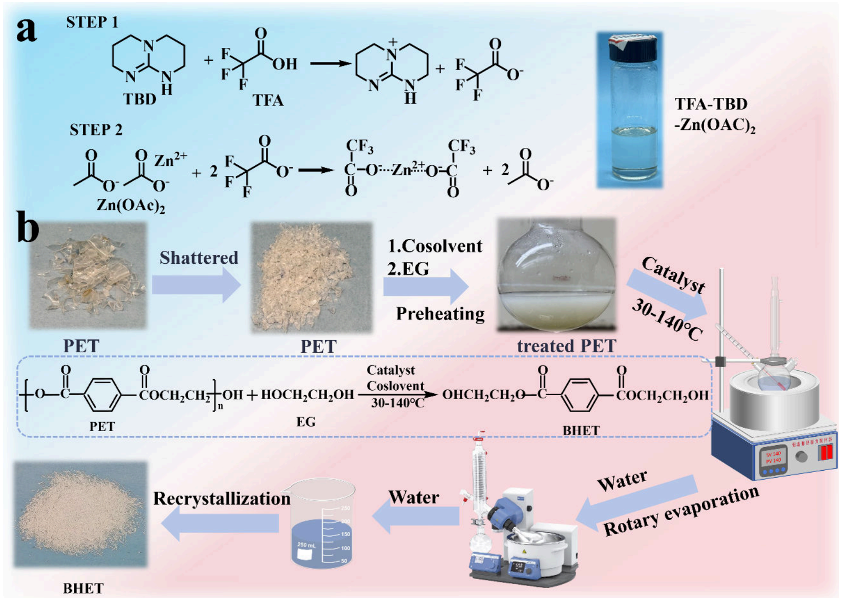

Figure 1. (a) Preparation of TFA–TBD–$Zn(OAc)_2$. (b) Reaction scheme and the reaction equation for glycolysis assisted by the co-solvent.

Table 1. Coded Levels of Independent Variables

factor	independent variables	coded level		
A	temperature (°C)	30	40	50
B	time (min)	60	90	120
C	mass ratio of EG to PET	3	4	5

were collected on filter paper after filtration and dried at 70 °C for 24 hours until constant weight was achieved, with the mass recorded as m_{BHET}. In addition, the co-solvent, EG, and catalyst were recycled by rotary evaporation at 80 °C. The conversion of PET and the yield of BHET were calculated as follows:

$$\text{Conversion of PET} = (m_{0,\,PET} - m_{1,\,PET})/m_{0,\,PET} \tag{1}$$

$$\text{Yield of BHET} = (m_{BHET}/M_{BHET})/(m_{0,\,PET}/M_{PET}) \tag{2}$$

where M_{PET} represents the molecular weight of the repeating unit of PET, and M_{BHET} represents the molecular weight of BHET.

2.5. Response Surface Methodology of the Reaction Conditions for Glycolysis at Low Temperature

Response surface methodology (RSM) with a Box-Behnken design model was applied to establish the quantitative relationship between BHET yields and three independent variables: (1) temperature, (2) time, and (3) mass ratio of EG to PET. This approach allows for comprehensive optimization of the reaction conditions. Based on preliminary experimental results, the investigated ranges for each independent variable were determined as follows (Table 1):

2.6. Characterization

The nuclear magnetic resonance hydrogen spectra (^{1}H-NMR) were measured by a Bruker AVANCE III HD 600M superconducting nuclear magnetic resonance spectrometer at room temperature with tetramethylsilane as the internal standard and deuteroxide or deuterium-chloroform as the solvent. Fourier transform infrared spectrum (FT-IR) analysis of the samples was used with the potassium bromide tablet compression method. The scanning range was 400−4000 cm^{-1}, and the scanning times were 16. A NETZSCH TG 209F3 thermogravimetric analyzer (TGA) was used to test the thermogravimetric behavior of the sample under the temperature rise rate of 20 °C/min and nitrogen atmosphere, and the test temperature range was 30−700 °C. The molecular weights of BEHT and treated PET were determined by ultra-high pressure liquid chromatogram-high resolution mass spectrometry (LC−MS). The BEHT was dissolved in methanol with a concentration of about 50−100 μg/mL. X-ray diffraction (XRD) patterns were collected using a Bruker X-ray diffractometer in Germany.

3. RESULTS AND DISCUSSION

3.1. Characterization of TFA–TBD–$Zn(OAc)_2$

The FT-IR characterization of synthesized TFA–TBD–$Zn(OAc)_2$ is presented in Figure 2a. 1786 cm^{-1} is the stretching vibration absorption peak of the carbonyl group −C=O of TFA, 3662 cm^{-1} is the stretching vibration absorption peak of the hydroxyl group −OH of TFA, 704 cm^{-1} belongs to the bending vibration absorption peak of C−F, and 1172 cm^{-1} is caused by the stretching vibration of C−O. For TBD, it can be known that 2357 cm^{-1} belongs to the stretching vibration absorption peak of C=N, 3650 and 1651 cm^{-1} correspond to the stretching vibration characteristic absorption peaks and bending vibration characteristic absorption peaks of the secondary amine N−H, and the absorption peak at 1438 cm^{-1} is caused by the stretching vibration of the C−N bond. The characteristic absorption peak at 2357 cm^{-1} is the characteristic absorption peak of the stretching vibration of C=N in TFA–TBD–$Zn(OAc)_2$. Obviously, the characteristic absorption peak of the hydroxyl group −OH at 3662 cm^{-1} of TFA disappears, and a new absorption peak appears at 3320 cm^{-1}. The absorption peaks at 1699 and 1408 cm^{-1} are caused by the blue shift of the −C=O absorption peak of TFA at 1786 cm^{-1} and the −C−N absorption peak of TBD at 1438 cm^{-1}, respectively. This indicates that an ionic bond is formed between the carboxyl group of TFA and the bridgehead nitrogen of TBD. The protonation of the ionic bond causes a change in the ring electron density, resulting in a blue shift of the characteristic absorption peaks of the C−N bond of TBD and the carbonyl group C=O of TFA. Additionally,

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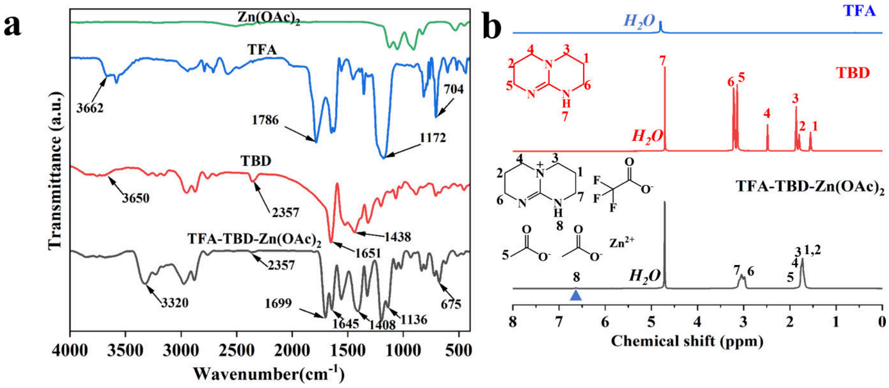

Figure 2. (a) FT-IR and (b) ^{1}H-NMR spectra of TFA−TBD−Zn(OAc)$_2$.

the characteristic absorption peak of the N−H bond appears to shift blue in TFA–TBD–Zn(OAc)$_2$. Obviously, due to the formation of hydrogen bonds between the N−H bond of TBD and the carbonyl carbon and carboxylate anion in the carboxylate group of TFA,[37] the stretching vibration absorption peak moves from 3650 to 3320 cm^{-1} and the bending vibration absorption peak moves from 1651 to 1408 cm^{-1}. In order to further demonstrate the formation of hydrogen bonds between the bifunctional catalyst, the ^{1}H-NMR spectra of TFA, TBD, and TFA–TBD–Zn(OAc)$_2$ were investigated (shown in Figure 2b). Due to the formation of ionic bonds between TFA–TBD–Zn(OAc)$_2$, the H−C−H proton absorption peak of TBD shifted from 2.48 to 1.76, and it overlapped with the 1,2,3 absorption peaks. Additionally, the formation of hydrogen bonds between the N−H bond of TBD and the carbonyl carbon and carboxylate anion in the carboxylate group results in a decrease in the electron cloud density around N−H, and the proton absorption peak shifted from 4.71 ppm to a downfield.[24] Therefore, a new proton absorption peak was observed at 6.64 ppm in TFA−TBD−Zn(OAc)$_2$.

3.2. Glycolysis of PET Treated with Co-solvents

The glycolysis of waste PET involves a mass transfer process of ethylene glycol,[38] so repeated crushing of the material facilitates increased contact area with the co-solvent to accelerate the treatment process. According to the principle of solubility parameters close to those of PET,[6] several co-solvents were selected.

The thermodynamic properties of solvents and polymers can be estimated using the well-established Hansen solubility parameters (HSPs),[39] which quantify the total cohesive energy density of a given component through three distinct parameters: δ_D, δ_P, δ_H, corresponding to the dispersion, polar, and hydrogen-bonding contributions to the HSP.[39] Among these, δ_D is associated with the dispersion cohesive energy arising from non-polar interactions, whereas δ_P is related to the polar cohesive energy coming from permanent dipole interactions. As for δ_H, this refers to the cohesive energy arising from hydrogen bonding. Although these parameters possess certain limitations—such as neglecting effects from induced dipoles, metallic bonding, or electrostatic interactions, and potential inaccuracies introduced when HSP values are derived empirically—previous studies have demonstrated their utility as a systematic screening tool for rational solvent selection. For instance, the "solubility parameter" Ra is employed to evaluate the affinity between a solvent and a polymer within the framework of the "sphere of solubility" concept, defined in eq 3:[40]

$$(R_a)^2 = 4(\delta_{D,\mathrm{PET}} - \delta_{D,2})^2 + (\delta_{P,\mathrm{PET}} - \delta_{P,2})^2 + (\delta_{H,\mathrm{PET}} - \delta_{H,2})^2 \quad (3)$$

where $\delta_{D,PET}$, $\delta_{P,PET}$, and $\delta_{H,PET}$ correspond to the dispersion, polar, and hydrogen Hansen solubility parameters (HSP) of PET (MPa$^{0.5}$). Essentially, a smaller value of Ra indicates stronger affinity and higher polymer-to-solvent intermingling.[39]

The HSPs of the selected co-solvents are summarized in Table 2 below. Meanwhile, the R_a values between these co-solvents and PET, calculated according to eq 3, are also presented in Table 2. As can be seen from the table, DMF and NMP exhibit the lowest R_a values with PET, indicating that these solvents possess superior affinity toward PET.

Furthermore, the effects of co-solvents on PET were further demonstrated through the following experiments. Then, these co-solvents were mixed with EG and then subjected to dissolution treatment of the waste PET flakes at a certain quality ratio under preheating conditions. The dissolution status is classified into four steps: (STEP1) completely insoluble, (STEP2) partially soluble, (STEP3) largely soluble, and (STEP4) completely soluble (as seen in Figure 3a) based on the experimental results. The schematic for co-solvent-assisted dissolution of waste PET is illustrated in Figure 3c. Moreover, Table 3 illustrates different co-solvents that were applied to the glycolysis of waste PET. Waste PET and EG with a mass ratio of 1:4 were mixed in a three-neck flask equipped with a condenser at 140°C for 1 h, using the TFA–TBD–Zn(OAc)$_2$ catalyst system at a loading of 5 wt % (relative to waste PET mass). After this, the effects of several co-solvents that can dissolve PET on the glycolysis process were investigated, and the obtained PET conversion and

Table 2. HSP and R_a Values of Several Co-solvents with PET

	δ_D	δ_P	δ_H	R_a
DMF	17.4	13.7	11.3	9.09
DMSO	18.4	16.4	10.2	10.71
NMP	18.0	12.3	7.2	6.18
DMI	18.0	15.0	9.0	9.11
PET	18.7	6.3	6.7	0

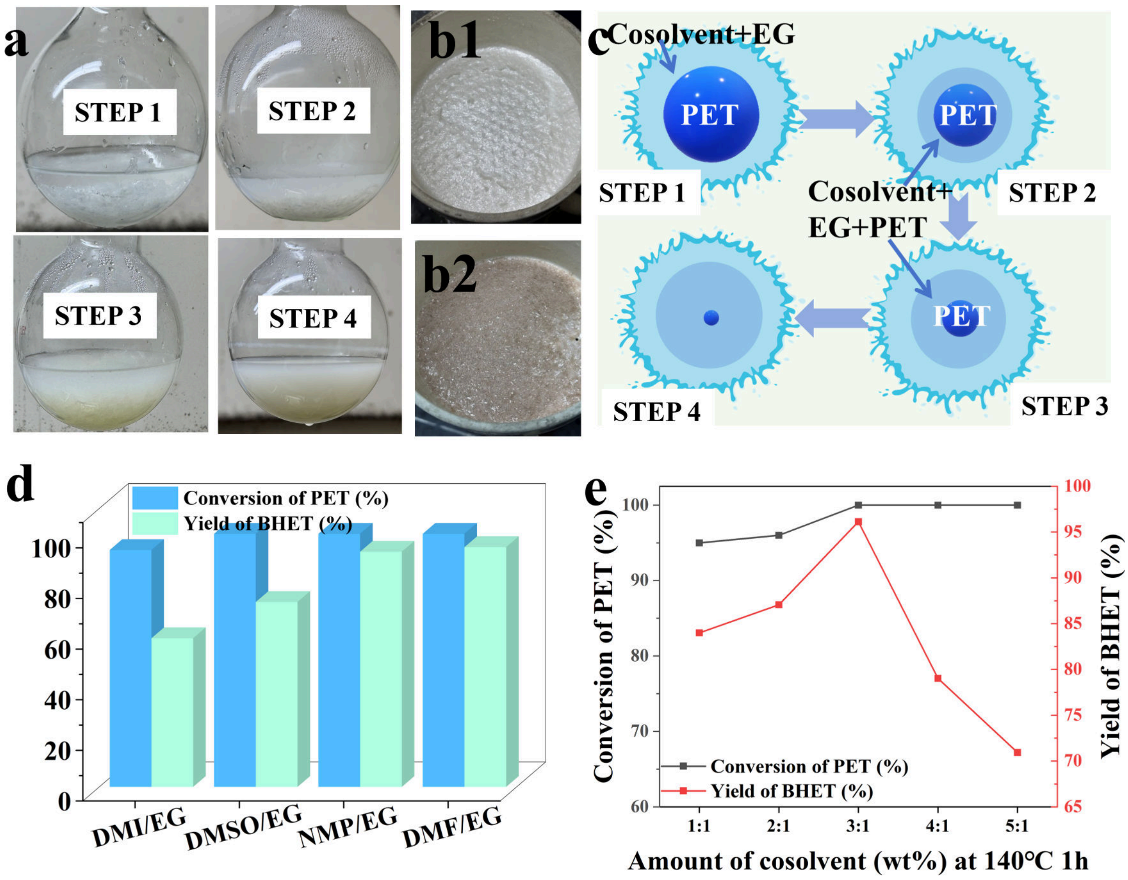

Figure 3. (a) Process of dissolving waste PET with a co-solvent. (b1) BHET obtained with DMF as the co-solvent. (b2) BHET obtained with NMP or DMSO as the co-solvent. (c) Schematic diagram of dissolving PET with the co-solvent and EG. (d) Effect of different co-solvents on the glycolysis of waste PET. (e) Influence of the mass ratio of the co-solvent to the waste PET on the glycolysis effect.

BHET yields are shown in Figure 3d. The research results show that NMP and DMF have effective dissolution and glycolysis effects on waste PET flakes. The conversion of PET reached 100%, and the yield of BHET was 92.9 and 94.7%, respectively. However, the by-products generated by the oxidative product of NMP have chromophores with conjugated double bonds, which caused the brownish-yellow color change of BHET[41] (as shown in Figure 3b2). Therefore, DMF was chosen as the subsequent co-solvent to assist the glycolysis of waste PET considering the simplicity of the purification of the BHET process. The excessive use of the co-solvent not only pollutes the environment and hinders product precipitation, but also raises production costs.[42] Therefore, we systematically investigated the influence of the mass ratio of DMF to PET on glycolysis efficiency. The conditions for the glycolysis reaction of waste PET were 140 °C for 1 hour, with a mass ratio of waste PET to EG of 1:4 and a catalyst TFA–TBD–Zn$(OAc)_2$ dosage of 7 wt % (shown in Figure 3e). When the amount of the co-solvent was equal to the amount of the waste PET, the conversion of PET and the yield of BHET were 95.1 and 84.0%, respectively. As the DMF-to-PET mass ratio increases, PET conversion reaches 100% under improved dissolution conditions. However, the excellent solubility of DMF toward PET, its oligomers, and the monomer BHET hampers BHET recrystallization; consequently, the optimal DMF-to-PET mass ratio is 3:1, where a PET conversion of 100% and a BHET yield of 95.7% were achieved simultaneously.

The waste PET treated with the co-solvent exhibited a white solid-state appearance, distinctly different from both pristine PET and the monomer BHET (Figure S1). To elucidate the influence of the co-solvent on waste PET, several approaches were employed to probe the underlying mechanism (shown in Figure 4). Figure 4a represents the FT-IR spectra of PET, BHET, and treated PET. The associative interaction of multiple hydroxyl groups led to the occurrence of a broad hydroxyl absorption peak in the treated PET at approximately 3400 cm^{-1}. Moreover, the other characteristic peaks indicate a long-chain polyol structure similar to that of BHET. Thermogravimetric analysis (TGA) was applied to investigate the thermal degradation profiles of

Table 3. Dissolution Step of Waste PET by Different Co-solvents[a]

	STEP 1	STEP 2	STEP 3	STEP 4
DMF/EG	√	√	√	√
DMI/EG	√	√	√	√
NMP/EG	√	√	√	√
DMSO/EG	√	√	√	√
THF/EG	×			
DCM/EG	√	√		
TCE/EG	√			

[a]The mass ratio of co-solvents to PET is 3:1.

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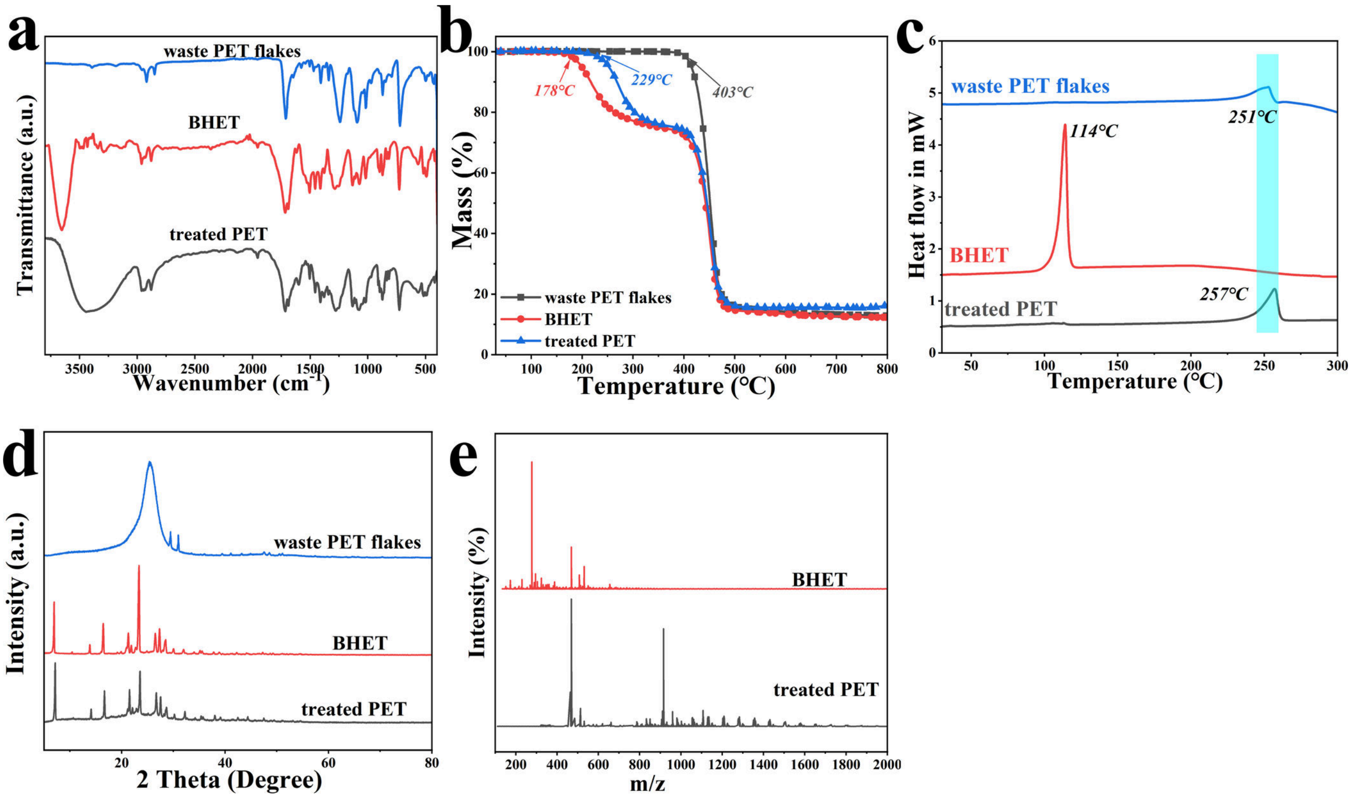

Figure 4. Comparative analysis of (a) FT-IR spectra, (b) TG curves, (c) DSC curves, (d) XRD curves, and (e) LC−MS spectra for PET, BHET, and treated PET.

treated PET, waste PET, and BHET (Figure 4b). The results show that waste PET begins to degrade at 403 °C, while the first degradation step of BHET monomers occurs at 178°C. Interestingly, treated PET exhibits its first degradation step at 229 °C. Its second degradation step occurred at 389 °C in accordance with BHET. As previously established,[43,44] the first thermal degradation step of BHET reflects decomposition of its dimer. Analogously, the initial mass-loss event in the treated PET is ascribed to thermal decomposition of the oligomer, whereas the second stage arises from scission of the PET backbone. The red curve in Figure 4c depicts a crystallization melting peak of waste PET flakes at 251 °C. BHET exhibits a melting peak at 114 °C, whereas the treated PET presents a melting peak at 257 °C. This indicates that co-solvent treatment alters the crystalline state of waste PET, enhancing its crystallization ability and resulting in a more ordered crystal structure. Consistent with the TGA data, the X-ray diffraction (XRD) pattern of treated PET (Figure 4d) confirms that the crystalline structure of waste PET has been modified.[45] Furthermore, the XRD pattern illustrated that the structure of treated PET presents an oligomer highly similar to that of BHET. This observation reinforces the previous DSC and TGA findings. Generally speaking, oligomers crystallize more readily than PET with a slightly higher melting temperature. To further characterize the molecular composition, treated PET was dissolved in a methanol/tetrahydrofuran mixture and analyzed via LC−MS (Figure 4e). The LC−MS results (Table S1) reveal that treated PET comprises a mixture of multiple long-chain oligomers, in comparison with the BHET spectrum. These findings demonstrate that the co-solvent pre-treatment is effective in disrupting intermolecular interactions within waste PET for cleaving its long molecular chains,[23,42] thereby facilitating PET depolymerization to achieve higher substrate conversion.

3.3. Influence of Catalyst Type and Amount, Temperature, and Time on the Glycolysis

To optimize the glycolysis of waste PET, the impacts of catalyst variety, content, and reaction conditions (time and temperature) were explored under constant co-solvent-to-PET (3:1) and EG-to-PET (4:1) mass ratios[46] (shown in Figure 5). As mentioned in our previous research,[43,44] the conversion of PET and the yield of BHET will be significantly improved through hydrogen bonding when the catalyst can enhance the nucleophilicity of hydroxyl groups in EG and the electrophilicity of carbonyl groups in repeating units of PETs. Based on this principle, the catalytic effect of the waste PET is shown in Figure 5a by studying different catalysts under the same conditions (5 wt % catalysts, 140 °C, and 1 hour). The recrystallization method of BHET at 0°C was used to estimate the yield of BHET. The solubility of BHET in DMF aqueous solution (containing 2 wt % DMF) was 25 wt % through recrystallization experiments. Consequently, the yields of BHET discussed later are all based on the modified values (dividing the calculated yield by 0.75). Zinc acetate ($Zn(OAc)_2$) was the first catalyst reported for the glycolysis of PET with a 100% PET conversion and a 90.1% BHET yield. Another bifunctional catalyst (TFA-DBU-$Zn(OAc)_2$) was prepared following the aforementioned experimental procedure to compare the catalytic effect of another organic base (DBU). Comparative studies on BHET yields during waste

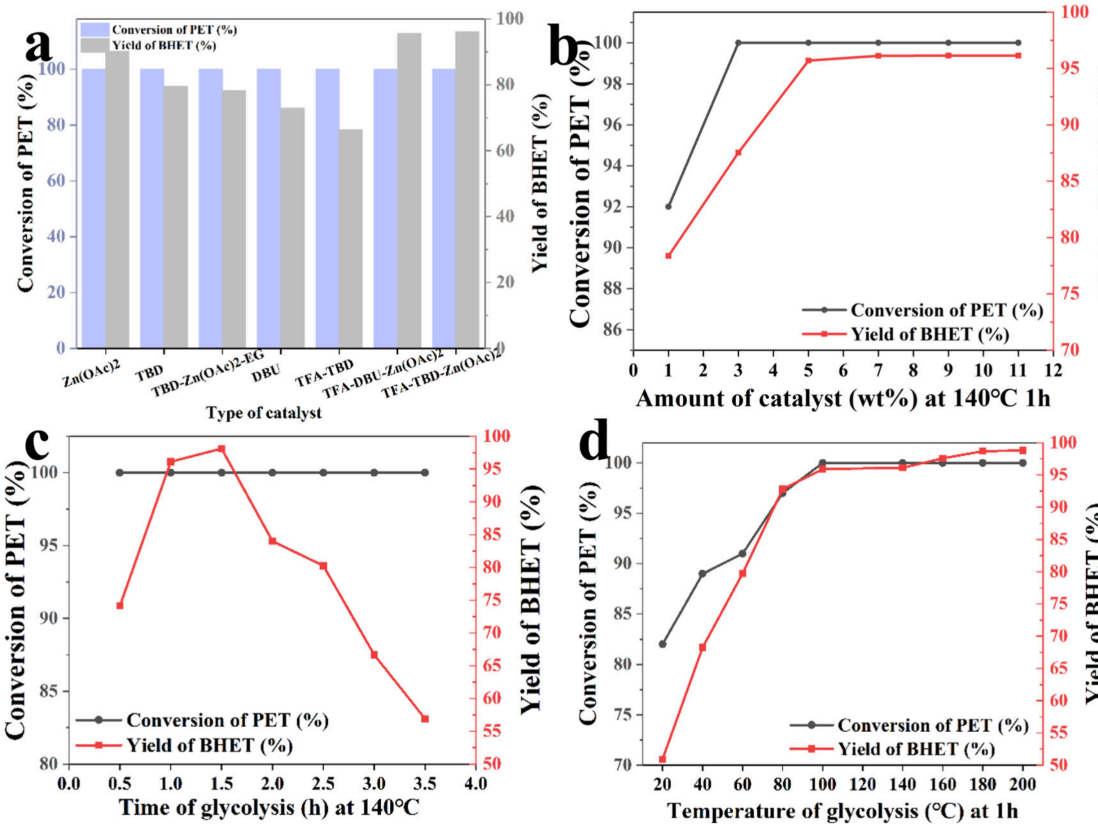

Figure 5. Influence of various factors on the glycolysis efficiency of waste PET: (a) type of catalysts, (b) amount of catalysts, (c) time of glycolysis, and (d) temperature of glycolysis.

PET glycolysis revealed that TFA–TBD–$Zn(OAc)_2$ demonstrated significantly enhanced catalytic efficiency. The bifunctional catalytic (TFA–TBD–$Zn(OAc)_2$) has an excellent performance with a 100.0% PET conversion and a 95.7% BHET yield, superior to $Zn(OAc)_2$, TBD, or TFA–TBD. This indicates that, through the synergistic interplay between $Zn(OAc)_2$ and TFA–TBD, the ternary catalyst TFA–TBD–$Zn(OAc)_2$ can fully exploit the combined Lewis acidity of Zn^{2+} and the nucleophilicity of OAc^-, thereby delivering markedly superior performance in the glycolysis of PET. TFA–TBD–$Zn(OAc)_2$ was selected as the catalyst in the subsequent research, and the influence of the catalyst dosage on the glycolysis efficiency of waste PET was investigated under the conditions of 140 °C and 1 hour (shown in Figure 5b). Obviously, both the conversion of PET and the yield of BHET continuously increase as the catalyst dosage increases. The yield of BHET reaches the maximum value of 95.9% at 7 wt % TFA–TBD–$Zn(OAc)_2$. The yield of BHET was not significantly increased as the catalyst content further increased. Therefore, the optimal dosage of TFA–TBD–$Zn(OAc)_2$ as a catalyst for catalyzing the glycolysis is 7 wt %. Next, we studied the glycolysis efficiency of waste PET at 140 °C for different durations (with a catalyst dosage of 7 wt %). As shown in Figure 5c, the waste PET was completely converted with a conversion reaching 100% at 140 °C due to the assistance of the co-solvent DMF. However, at the initial 0.5 hours, the yield of BHET was only 74.1%. As the glycolysis time prolonged, the yield of BHET reached maximum 96.1% at 1.5 hours. However, prolonged reaction time leads to a continuous decline in the BHET yield due to partial dimerization of BHET into its dimer at this temperature, which negatively impacts the overall BHET production.[47,48] Given that temperature plays a crucial role in waste PET glycolysis efficiency, the reaction was evaluated under controlled conditions: 5 wt % catalyst loading and 1 hour reaction time. As illustrated in Figure 5d, PET conversion increases with temperature, attaining complete conversion (100.0%) at 100 °C. Concurrently, the BHET yield exhibits a steady upward trend, reaching a maximum of 98.7% at 180°C. Although higher temperatures could potentially further improve the BHET yield, the substantial energy input required would contradict the principles of green and efficient glycolysis. Consequently, the optimal reaction conditions—300wt % co-solvent and 7 wt % catalyst loading at 140 °C for 1.5 hours—deliver both full PET conversion (100.0%) and an exceptional BHET yield of 96.1%.

Interestingly, the results showed that, the conversion of PET and the yield of BHET exceeded the expected values (89.2 and 68.0%) at around room temperature, which was not reported in previous research. We hypothesized that if the glycolysis time at room temperature was further prolonged, the conversion of PET and the yield of BHET would reach the ideal results for the green and efficient chemical recycling of waste PET. Therefore, glycolysis reaction was further examined at 30 °C under varying reaction times, with an EG/waste PET mass ratio of 4:1, 7 wt % catalyst loading and a co-solvent/waste PET mass ratio of 3:1. As Figure 6a presents, the conversion of PET reached 90.0% with a BHET yield of 35.3% within the first 0.5 hours. The conversion progressively increased to

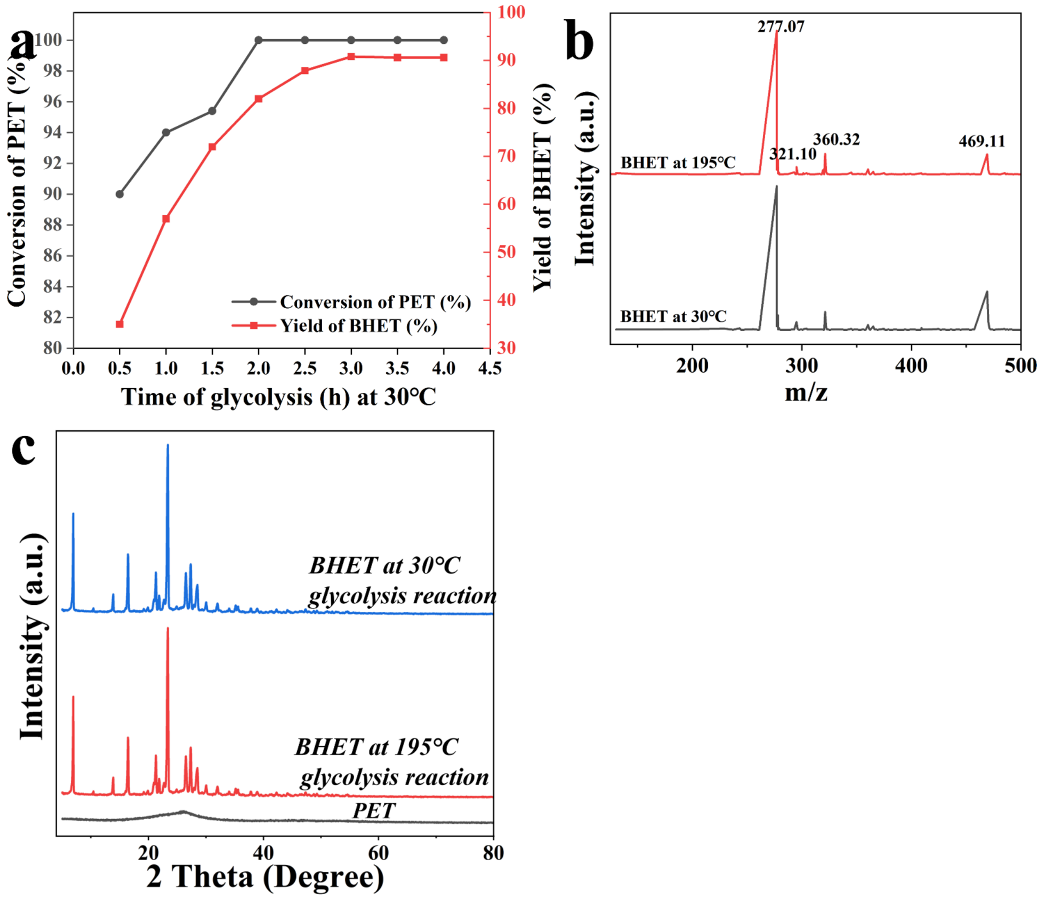

Figure 6. (a) Influence of different glycolysis time at 30 °C on the conversion of PET and the yield of BHET. The comparison of the (b) LC−MS and (c) XRD spectra of BHET obtained by glycolysis at 30 °C and 195 °C.

100.0% after 2 hours, while the BHET yield attained 90.8% at 3 hours. Then, prolonging the reaction time beyond this point did not significantly improve the yield of BHET with extended duration, indicating that low-temperature glycolysis achieves near-complete depolymerization. To verify consistency of the product at different temperatures, the molecular weight of BHET obtained at 30 °C and 195 °C was analyzed by HPLC (Figure 6b). The dominant peak corresponding to an m/z of 277.06 confirmed the presence of BHET monomers bound to Na^+ ions and demonstrated identical molecular weights. XRD analysis (Figure 6c) further confirmed that BHET produced at both temperatures shares the same crystalline structure, validating structural uniformity. This study successfully demonstrates the feasibility of low-temperature glycolysis (30 °C). Compared to conventional processing at 140 °C (1 hour), this method requires only a moderately longer reaction time (3 hours) while substantially reducing energy input, aligning with the principles of green and sustainable chemical recycling.

3.4. Glycolysis Conditions at Low Temperature

To provide more professional guidance for the implementation of chemical recycling in the chemical industry, the critical parameters in the glycolysis reaction (such as temperature, time, and the mass ratio of EG to PET) still need to be systematically studied for the feasibility of glycolysis for waste PET under low-temperature conditions. Based on the Box-Behnken model and the independent variable results in Table 1, we designed the following 17 experimental groups (with a catalyst content of 7 wt % and a ratio of co-solvent to PET mass being 3:1). The experimental results of the waste PET under different

Table 4. Yield of BHET under Different Factors[a]

	A	B	C	yield of BHET (%)
1	40	60	3	68.67
2	40	120	5	85.13
3	50	60	4	84.07
4	40	90	4	81.47
5	50	90	5	88.07
6	30	120	4	82.80
7	40	90	4	81.47
8	50	120	4	90.80
9	40	90	4	81.47
10	30	90	5	71.73
11	30	90	3	72.53
12	40	90	4	81.47
13	50	90	3	88.80
14	40	120	3	86.80
15	30	60	4	57.47
16	40	90	4	81.47
17	40	60	5	67.33

[a]Glycolysis condition: 5 g of PET wastes, 0.35 g of TFA–TBD–$Zn(OAc)_2$ catalysts, and 15 g of the co-solvent DMF.

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levels are shown in Table 4. Obviously, Table 4 demonstrates the significant changes in the yield of BHET under different parameter values, and these results emphasize the effectiveness of parameter selection and specified ranges within the experimental design framework.

Based on the data in Table 4, Design Expert software (Stat-Ease, Inc.) was used to calculate the RSM regression model and conduct variance analysis (ANOVA). The coefficient evaluation, significance test, and statistical results of the response surface model are shown in Table 5 below. The actual quadratic regression equation obtained is

$$\begin{aligned}\text{Yield of BHET} = &-65.80 + 2.06 \times A + 1.49 \times B \\ &+ 5.57 \times C - 0.01 \times A \times B + 0.0017 \times A \times C - 0.0027 \\ &\times B \times C - 0.0044 \times A^2 - 0.0042 \times B^2 - 0.75 \times C^2 \end{aligned} \tag{4}$$

The p-value of the model is less than 0.0500, and the F-value of 4.41 indicates that the model terms are significant, from the ANOVA results of the fitted model. In this case, A and B are important model terms. Adequate Precision (Adeq Precision) is an important metric in ANOVA output for evaluating the signal-to-noise ratio of the model and reflects the reliability of its predictive capability. The Adeq Precision of this model is 7.74 (>4) with excellent predictive capability and trustworthy optimization results. Figure 7a demonstrates excellent agreement between the predicted values of the model and experimental data, with both datasets following the fitted function distribution. The close agreement between predicted and experimental values across multiple runs confirms the accuracy of the model, as reflected by the high R^2 value. Therefore, this model effectively captures the relationship between the independent variables and the observed response for design space exploration. As seen in Figure 7b−d, the response surface analysis reveals that the optimal conditions are an EG/waste PET mass ratio of 4, a reaction temperature of 50 °C, and a reaction time of 2 hours when TFA–TBD–$Zn(OAc)_2$ is used as the catalyst. Within the design space, the use of TFA–TBD–$Zn(OAc)_2$ with DMF as a co-solvent enables a high BHET yield of 90.8% under low-temperature catalytic depolymerization of waste PET. In comparison to the data reported in the literature (Table 6), the glycolysis of PET in this study was conducted at significantly lower temperatures while achieving a higher yield of BHET. This finding establishes co-solvent-assisted glycolysis as a viable strategy for sustainable PET recycling, offering the combined advantages of milder reaction conditions and enhanced monomer recovery.

3.5. Kinetic of Glycolysis Catalyzed by TFA–TBD–$Zn(OAc)_2$

The investigation demonstrates that using a 7 wt % TFA–TBD–ZnAc catalyst in a DMF-enhanced glycolysis system (mass ratios of the co-solvent and EG to PET at 3:1 and 4:1, respectively) enables complete PET conversion under two distinct regimes: a high-temperature rapid path (140 °C, 1.5 h) resulting in 96.1% BHET and a low-temperature path (30 °C, 3 h) achieving 90.8% BHET.

In previous studies,[38,49] it was known that the depolymerization reaction of PET is generally a first-order reaction. Therefore, the reaction of degrading waste PET flakes into BHET was assumed a first-order reaction. Then, the linear correlation coefficient was used to verify the consistency between the reaction and the first-order reaction kinetics model. The reaction rate constant is typically correlated with the conversion of the reactants (specifically, the conversion of PET). However, the waste PET in this case has been fully dissolved in the co-solvent system due to the presence of co-solvents, which makes accurate determination of PET conversion particularly challenging. Thus, the BHET yield is used as an experimental parameter in this formula to study the reaction kinetics, and the first-order reaction kinetics is as follows:

$$\ln \frac{1}{1-Y} = k \times t \tag{5}$$

where Y, k, and t represent the yield of BHET, the reaction rate constant, and the reaction time, respectively. Moreover, we set four experimental temperatures (30, 40, 50, and 60 °C) and time (60, 90, 120, and 150 min) to investigate the yield of BHET (as shown in Figure 8a). Figure 8b depicts a graph of $\ln(1/(1-Y))$ versus reaction time, and the slope obtained from linear fitting is the corresponding reaction rate constant at this temperature (Table 7). The reaction rate constants for PET degradation between 30 and 60 °C exhibit a positive correlation with temperature, indicating that elevated temperatures promote the reaction. This observation is consistent with prior studies on the effect of temperature on PET conversion and BHET yields. The high linear correlation coefficients (R^2 > 0.93, Table 7) confirm the validity of the first-order reaction kinetics model for this system.

Arrhenius formula 6 was applied for calculating the reaction apparent activation energy (E_a) of the reaction

$$\ln k = \ln A - E_a/RT \tag{6}$$

where R is the gas constant (8.314 J × K^{-1} × mol^{-1}), T is the reaction temperature in Kelvin, A is the pre-exponential factor, and k is the reaction rate constant. The Arrhenius plot (Figure 8c) was generated by correlating the reaction rate constant with temperature, demonstrating the temperature dependence of the reaction kinetics. Based on the Arrhenius plot of the reaction rate constant, the equation obtained through fitting is

$$y = -7754.29x + 21.23 \tag{7}$$

The slope of the linear regression equation is −7754.29, and R^2 is 0.99. The E_a of the PET glycolysis reaction catalyzed by TFA–TBD–$Zn(OAc)_2$ is 64.47 kJ/mol, which further substan-

Table 5. Evaluation of Coefficients and Significance Test of the RSM Model

source	P-value	significance
model	<0.0001	significant
A	<0.0001	significant
B	<0.0001	significant
C	0.2059	not significant
$A*B$	<0.0001	significant
$A*C$	0.9578	not significant
$B*C$	0.8034	not significant
A^2	0.1978	significant
B^2	<0.0001	significant
C^2	0.0478	not significant
R^2	0.99	significant

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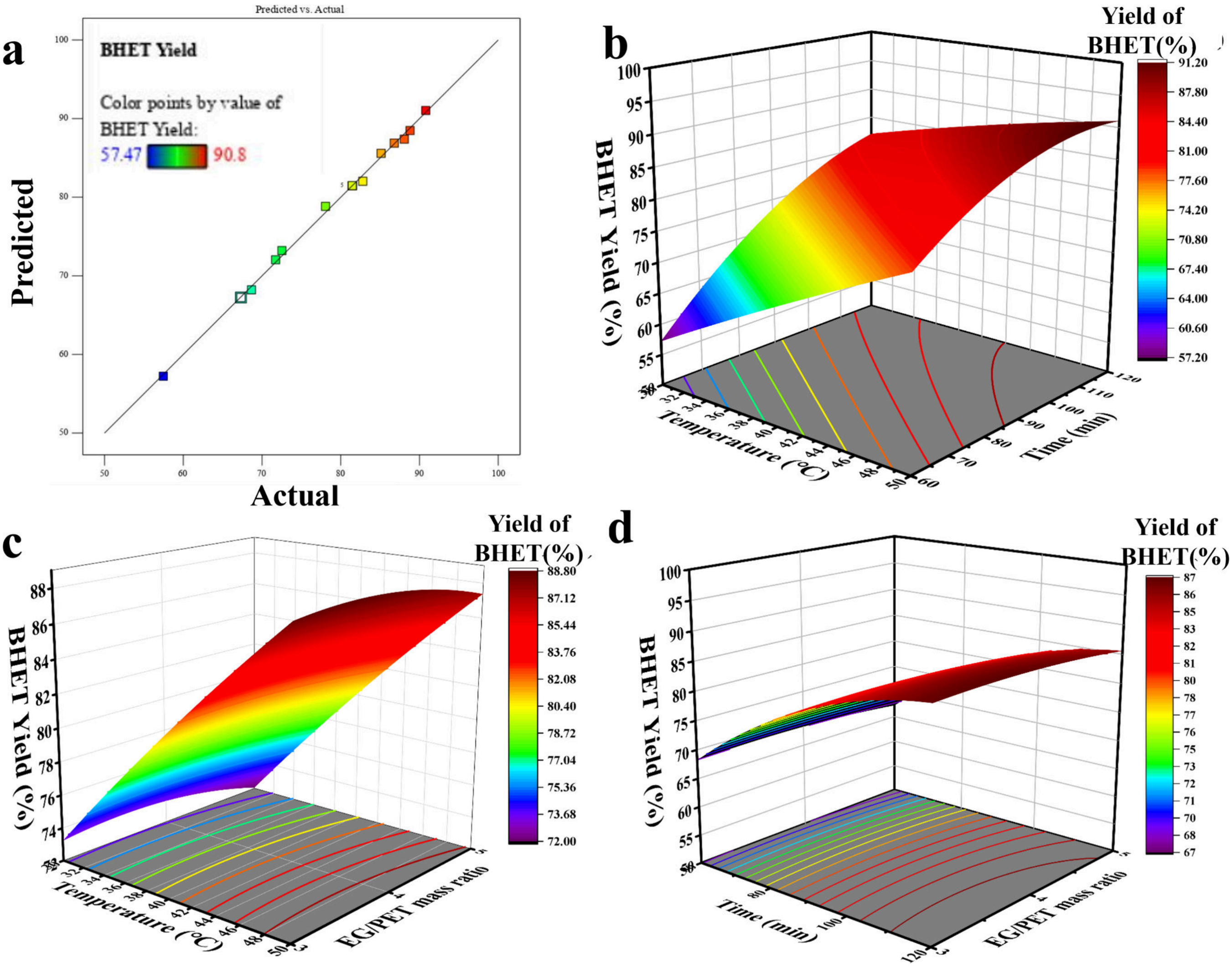

Figure 7. RSM results of glycolysis at low temperatures: (a) actual values vs. predicted values, (b) yield of BHET under different temperatures and times, (c) yield of BHET under different temperatures and the mass ratio of EG and PET, and (d) yield of BHET under different times and the mass ratio of EG and PET.

tiates the enhancing effect of the catalyst and co-solvent on PET glycolysis.

3.6. Proposed Mechanism of PET Glycolysis Catalyzed by TFA–TBD–Zn(OAc)$_2$

Our previous studies demonstrated that catalysts capable of forming intermolecular hydrogen bonds with ethylene glycol (EG) and modulating the O−H bond length in EG can enhance nucleophilic attack by the EG hydroxyl group on the ester carbonyl of repeating units of PETs, thereby accelerating the glycolysis reaction. As demonstrated earlier, TFA–TBD–Zn(OAc)$_2$ can form both ionic and hydrogen bonds intrinsically, while its nitrogen atom further facilitates hydrogen bonding with the hydroxyl group (O−H) of EG. The ^{1}H-NMR spectra of EG with different molar ratios (1,0, 0:1, and 1:1) of the catalyst TFA–TBD–Zn(OAc)$_2$ are shown in Figure 9a. As shown in the figure, the characteristic absorption peak of DMSO appears at 2.50 ppm. The methylene protons (−CH$_2$−) of ethylene glycol (EG) resonate at 3.39 ppm, exhibiting a doublet splitting pattern due to coupling with the adjacent hydroxyl proton (−O−H). The hydroxyl proton (−O−H) of EG itself gives rise to a signal at 4.45 ppm. Additionally, the proton of the −N−H group in TFA–TBD–Zn(OAc)$_2$ is observed at 7.95 ppm. Upon mixing the catalyst TFA–TBD–Zn(OAc)$_2$ with EG in a 1:1 molar ratio, the proton resonance of the −O−H in EG shifted downfield to 4.52 ppm, while the −N−H

Table 6. Comparison of Reaction Temperature and Yields of BHET

catalyst	temperature of glycolysis (°C)	yield of BHET (%)	article
[Bmim]$_2$[CoCl$_4$]	175	82	7
TBD	65	78	23
ChCl/Zn(OAc)$_2$	90	90	29
TBD/TBuOK	100	90	31
[Ch][Gly]	150	51	49
Pd-Cu/γ-Al$_2$O$_3$	160	86	50
TFA–TBD–Zn(OAc)$_2$	30	91	this work

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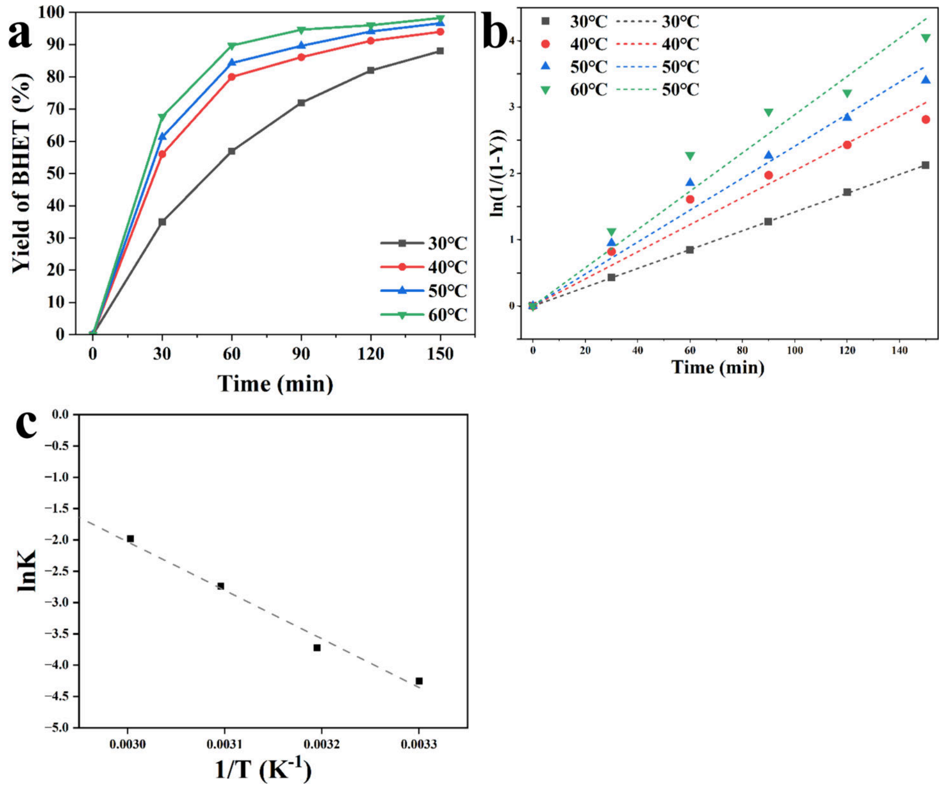

Figure 8. (a) Yield of BHET at different reaction time and temperatures, (b) fitted first-order reaction kinetics equation, and (c) fitted Arrhenius equation.

proton signal of TFA–TBD–Zn(OAc)$_2$ shifted upfield to 7.80 ppm. These chemical shift changes confirm the formation of hydrogen-bonding interactions between the catalyst and EG. The proposed interaction is illustrated in Figure 9b.

Based on the above findings, the glycolysis mechanism of waste PET flakes catalyzed by TFA–TBD–Zn(OAc)$_2$ in the presence of a co-solvent can be proposed (Figure 9c). The co-solvent first facilitates the dissolution of PET flakes in the mixed solvent system containing EG. Subsequently, the carbonyl oxygen of the PET ester group is coordinated by the Zn^{2+} ion in the catalyst with resulting polarization of the C=O bond along with increased electrophilicity of the carbonyl carbon and enhanced nucleophilicity of the carbonyl oxygen. Simultaneously, the TBD cation establishes intermolecular hydrogen bonds with EG, which activates the hydroxyl group and enhances its nucleophilic attack on PET's electrophilic carbonyl carbon. Consequently, the cleavage of the ester bond produces oligomers with a range of molecular weights. These oligomers progressively depolymerize to yield BHET dimers and ultimately the monomeric BHET product. Therefore, it is precisely this synergistic interplay between the Lewis acid and Lewis base sites that substantially reduces the activation energy of the reaction, thereby enabling rapid glycolysis of waste PET and facilitating enhanced BHET yields under mild temperature conditions.

Table 7. Fitted Rate Constants and Correlation Coefficients

temperature (°C)	rate constants (min^{-1})	R^2
30	0.01984	0.99
40	0.04968	0.95
50	0.11664	0.93
60	0.25881	0.96

4. CONCLUSIONS

This study established DMF/EG as an effective co-solvent system for the dissolution of PET and developed a novel bifunctional catalyst (TFA–TBD–Zn(OAc)$_2$) for the glycolysis of PET. Response surface methodology analysis revealed that temperature and reaction time are the critical factors governing the glycolysis of PET at low temperatures. The optimal conditions for glycolysis were determined to be a co-solvent-to-PET mass ratio of 3:1 and a catalyst loading of 7 wt %. Complete depolymerization of waste PET materials was achieved, with the system delivering a yield of BHET of 96.1% within 1.5 h at 140 °C and maintaining a high yield of 90.8% even at a reduced temperature of 30 °C for 3 h. Remarkably, the synergistic effect between the co-solvent and bifunctional catalyst enabled the glycolysis of PET to proceed effectively at near-ambient temperature. Kinetic studies confirmed that the catalyzed glycolysis follows first-order reaction kinetics with an activation energy of 64.47 kJ/mol. ^{1}H-NMR analysis demonstrated that the catalytic mechanism involves two key interactions, whereby hydrogen bonding between TFA–TBD–Zn(OAc)$_2$ and EG enhances the nucleophilicity

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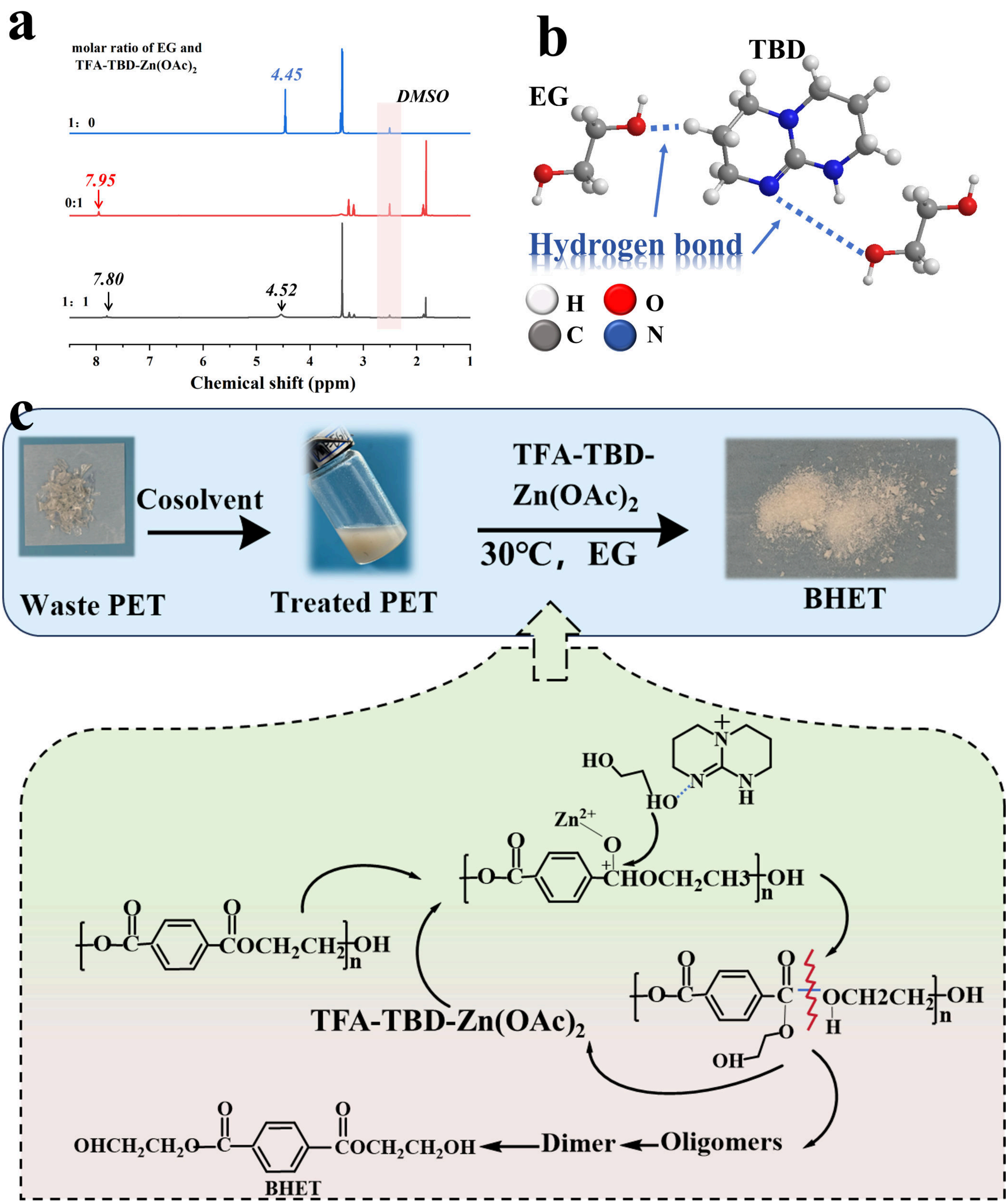

Figure 9. (a) [1]H-NMR of EG and TFA–TBD–Zn(OAc)$_2$, (b) hydrogen-bond interaction between EG and TFA–TBD–Zn(OAc)$_2$, and (c) mechanism diagram of the glycolysis reaction.

of the hydroxyl oxygen in EG and Zn^{2+} coordination with the PET ester group increases the electrophilicity of the carbonyl carbon, thereby collectively accelerating the depolymerization of PET. These findings provide valuable insights for developing processes for PET glycolysis that are more efficient and sustainable at low temperatures. Future research should focus on the depolymerization of mixed PET products at room temperature with varying crystallinity and molecular weights, which

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ACS Sustain. Resour. Manag. 2026, 3, 1155–1168

would significantly expand the range of sources of PET waste that can be processed while minimizing energy consumption.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acssusresmgt.6c00011.

Photos of PET, treated PET after dried and BHET, and LC−MS data of waste PET after solvent treatment (DOCX)

AUTHOR INFORMATION

Corresponding Author

Hui He − *School of Materials Science and Engineering, South China University of Technology, Wushan Road, Tianhe District, Guangzhou, Guangdong 510640, China;* orcid.org/0000-0001-7351-2830; Email: pshuihe@scut.edu.cn

Authors

Cheng Zhang − *School of Materials Science and Engineering, South China University of Technology, Wushan Road, Tianhe District, Guangzhou, Guangdong 510640, China;* orcid.org/0000-0003-4367-6599

Fan Kang − *School of Materials Science and Engineering, South China University of Technology, Wushan Road, Tianhe District, Guangzhou, Guangdong 510640, China*

Hongyu Zhai − *School of Materials Science and Engineering, South China University of Technology, Wushan Road, Tianhe District, Guangzhou, Guangdong 510640, China;* orcid.org/0009-0006-6119-5683

Complete contact information is available at: https://pubs.acs.org/doi/10.1021/acssusresmgt.6c00011

Notes

The authors declare no competing financial interest.

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